

PROTOCOL: Agarose Gel Check

To confirm PCR amplification success from ONTOLOGY™ PCR Plates, completed reactions can be visualized using gel electrophoresis. This protocol uses the Invitrogen E-gel® 96 with SYBR Safe stain. For other electrophoresis approaches, ONTOLOGY™ development team (dev@ontology.bio) for advice.

🔗 Bullet 🔗 points show tips for executing the protocol.

(1) Loading and running E-gel® 96 gels

- 🔗 The recommended program for 2% Agarose E-gel® 96 gel is **EG** and the run time is **6-12 min.**
- a. Plug the Mother E-Base™ into an electrical outlet. Press and release the pwr/prg (power/program) button on the base to select program EG.
- b. Remove E-gel® from the package and remove plastic comb.
- c. Slide gel into the two electrode connections on the Mother or Daughter E-Base™.
- d. Load **16 µL** of molecular grade H₂O into wells with multichannel pipettor.
- e. Optional: Load appropriate DNA markers in the marker wells ("M" column on the right); e.g. 4 µL of Invitrogen E-Gel 1 Kb Plus DNA Ladder, #10488090.
- f. Load **4 µL** of sample.
- g. To begin electrophoresis, press and release the pwr/prg button on the E-Base™. The red light will change to green.
- h. At the end of run (signaled with a flashing red light and rapid beeping), press and release the pwr/prg button to stop the beeping.
- i. Remove gel cassette from the base and capture a digital image of a gel on UV transilluminator equipped with digital camera.
- j. Analyze the image and align or arrange lanes in the image using the E-editor™ 2.0 software available at: <http://www.invitrogen.com/egels/>



*Example E-gel® with successful amplification
for most samples*

